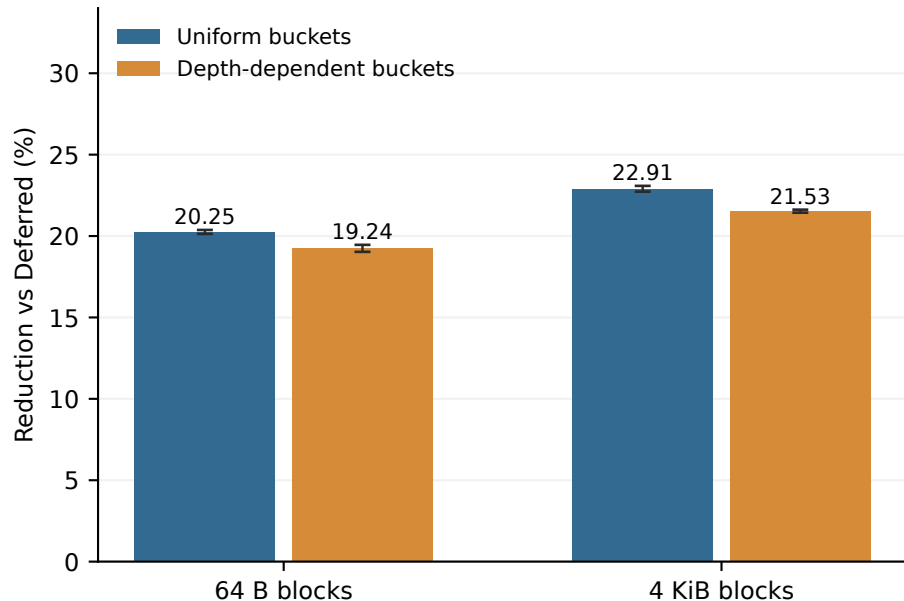
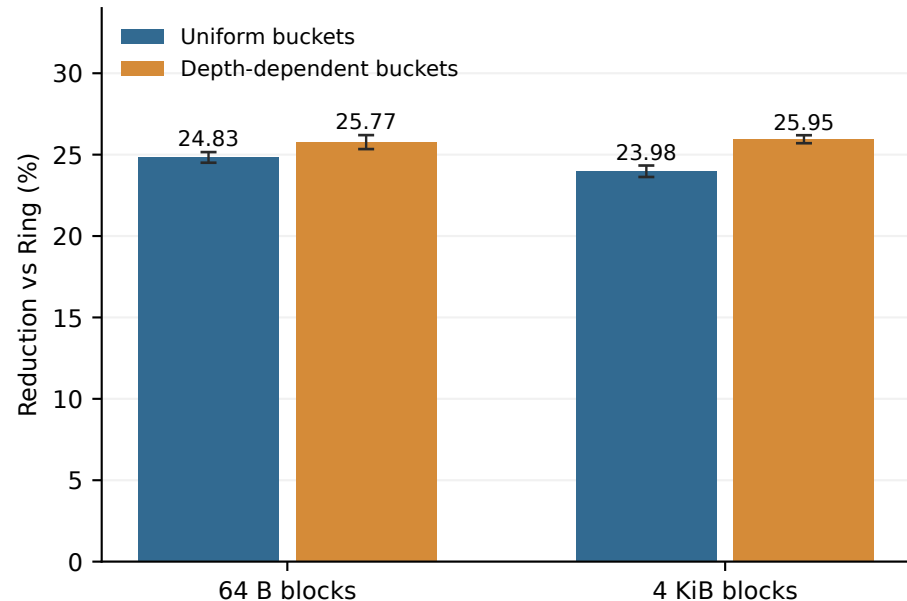


(a) Static IR allocation + prefix cache



(b) Static AB dummy layout + prefix cache



N = 4,096; uniform requests; five paired runs, mean and 95% t intervals.

2,048 warmup + 4,096 measured requests. IR: R480, six cached levels; AB: R256, five cached levels.

Static components only; no native IR-Stash/DWB or AB CB/DeadQ. All serialized protocol bytes included.